7.7 (a) Inside the medium j, at x. $E = E_+ e^{ik_j x} + E_- e^{-ik_j x}$, then at $x + t_j$,

$$E' = E_+ e^{ik_j(x+t_j)} + E_- e^{ik_j(x+t_j)} = E_+' e^{ik_j x} + E_-' e^{-ik_j x}.$$

Then $E_+' = E_+ e^{ik_j t_j}$, $E_-' = E_- e^{-ik_j t_j}$, or

$$\begin{pmatrix} E_+' \\ E_-' \end{pmatrix} = \begin{pmatrix} e^{ik_j t_j} & 0 \\ 0 & e^{-ik_j t_j} \end{pmatrix} \begin{pmatrix} E_+ \\ E_- \end{pmatrix}.$$

and $T_{layer}(n_j, t_j) = \begin{pmatrix} e^{ik_j t_j} & 0 \\ 0 & e^{-ik_j t_j} \end{pmatrix} = I \cos(k_j t_j) + i\sigma_3 \sin(k_j t_j)$,

with $k_j = n_j \omega / c$.

(b) At the surface, we have

$$E = E_+ e^{ik_1 x} + E_- e^{-ik_1 x}, \quad x < 0$$

$$E = E_+' e^{ik_2 x} + E_-' e^{-ik_2 x}, \quad x > 0$$

and the boundary condition is

$$E_+ + E_- = E_+' + E_-'$$

$$n_1 (E_+ - E_-) = n_2 (E_+' - E_-')$$

Let $n = n_1/n_2$, it can be shown that

$$E_+' = \tfrac{1}{2}(1+n) E_+ + \tfrac{1}{2}(1-n) E_-,$$

$$E_-' = \tfrac{1}{2}(1-n) E_+ + \tfrac{1}{2}(1+n) E_-$$

Therefore, $T_{interface}(2,1) = \frac{1}{2}\begin{pmatrix} 1+n & 1-n \\ 1-n & 1+n \end{pmatrix} = I\,\frac{1+n}{2} + \sigma_1 \frac{1-n}{2}$

(c) Given the transfer matrix,

$$\begin{pmatrix} E_+' \\ E_-' \end{pmatrix} = T \begin{pmatrix} E_+ \\ E_- \end{pmatrix} = \begin{pmatrix} t_{11} E_+ + t_{12} E_- \\ t_{21} E_+ + t_{22} E_- \end{pmatrix}$$

Since $E_-' = 0$, we have $t_{21} E_+ + t_{22} E_- = 0$, or $E_- = -\frac{t_{21}}{t_{22}} E_+$.

On the other hand, $E_+' = t_{11} E_+ + t_{12} E_- = \left(t_{11} - \frac{t_{12} t_{21}}{t_{22}}\right) E_+ = \frac{\det(T)}{t_{22}} E_+$.

Therefore. $E_{trans} = \frac{\det(T)}{t_{22}} E_{inc}$, $E_{refl} = -\frac{t_{21}}{t_{22}} E_{inc}$